Network Cloud Design
Project 1 Presentation

Routing with OSPF and configure VLANs with VLSM

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- Testing & Troubleshooting
- Network Recommendations
- Final Demonstration

Project Summary & Requirements

Project Summary & Requirements

Summary of the Project:

- XYZ Research Company has hired us to help design their network
- A formal report is to be presented after
- Logical diagrams of the network and physical topology to be submitted
- Prototype of the network should also be built for verification purposes

Project Summary & Requirements

Background of the client (XYZ Research):

- XYZ Research is a small company that's developing high-speed wireless products
- Comprises of the Main Building, Admin Building, and Remote Sales Office
- Multiple employee groups (Administration, Sales & Marketing, Research & Development)
- Seeking to implement a wired network that supports 100% growth over the next 5-10 years

Project Summary & Requirements

XYZ Research Company Requirements:

- 18 employees in the Research and Development group
- 9 employees in the Sales and Marketing group
- 7 employees in the Administration group
- 5 employees in the remote sales office
- Lifetime Max of 5 servers on separate subnet, regardless of company growth

Project Summary & Requirements

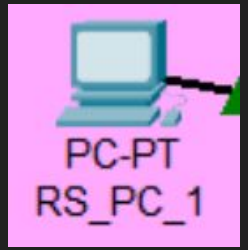
XYZ Research Company Requirements (Continued):

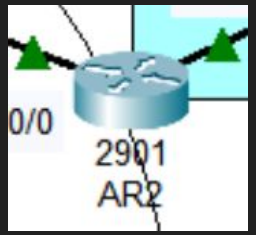
- Use subnet 200.200.100.0/30 for connection to the Internet router
- Use public class C network 220.0.1.0/24 for internal addressing
- Use VLSM for IP addressing
- Expect 100% growth of current IP requirements when determining size of subnets
- All networking devices must have IP addresses

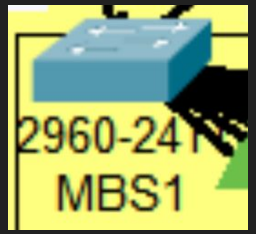
Logical Topology Diagram

Naming Conventions

- End Devices
  - X**PC**Y
    - X: Network Name Summary (e.g. RS for Remote Sales LAN)
    - Y: PC Number (e.g. RS_PC_1 for first PC in Admin LAN)
- Routers
  - X**R**Y
    - X: Location of router (e.g. A for Admin Building)
    - Y: Router number (e.g. AR2 for Admin Building Router 2)
- Switches
  - X**S**Y
    - X: Location of switch (e.g. MB for Main Building)
    - Y: Switch number (e.g. MBS1 for Main Building Floor 1)

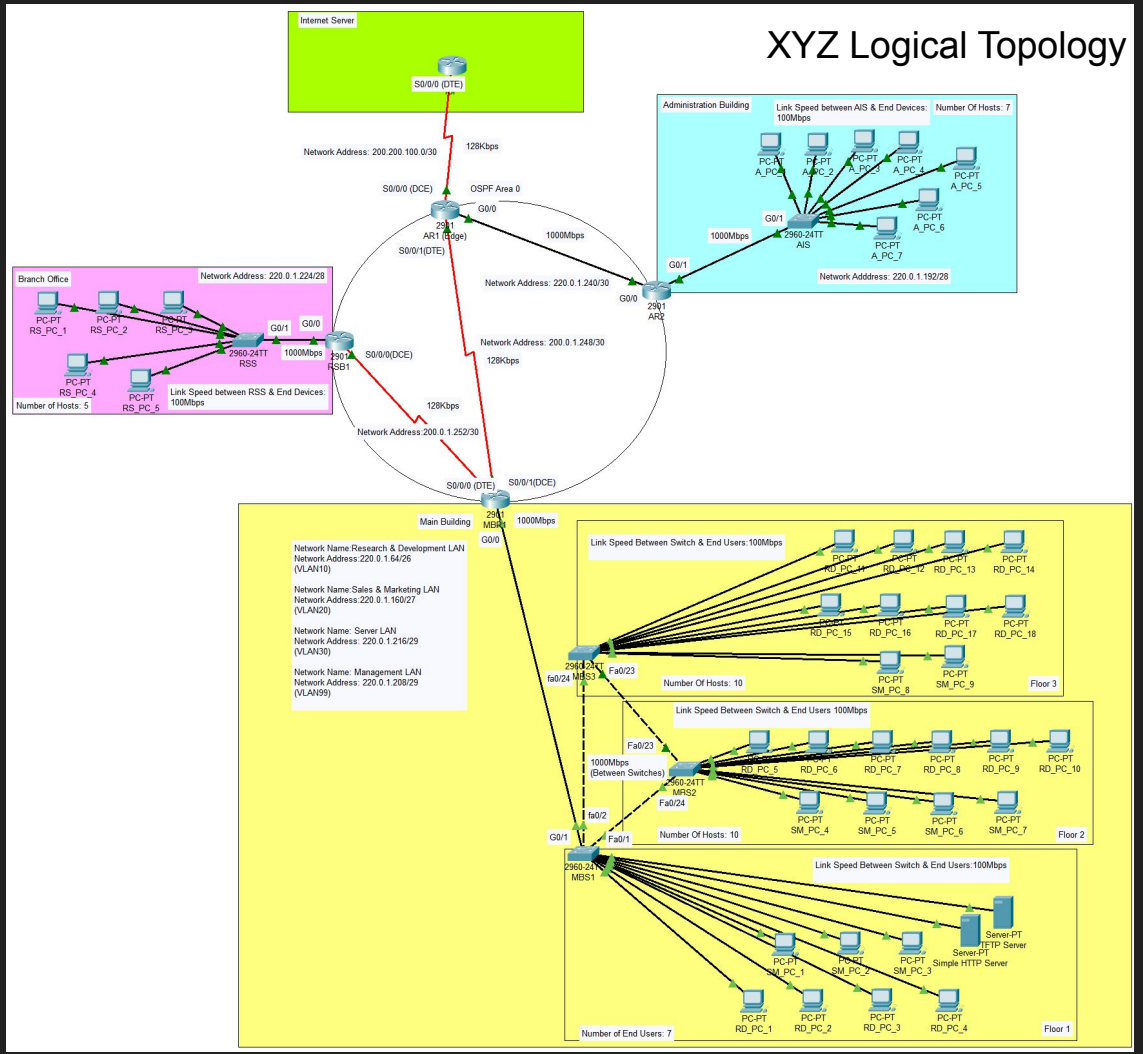






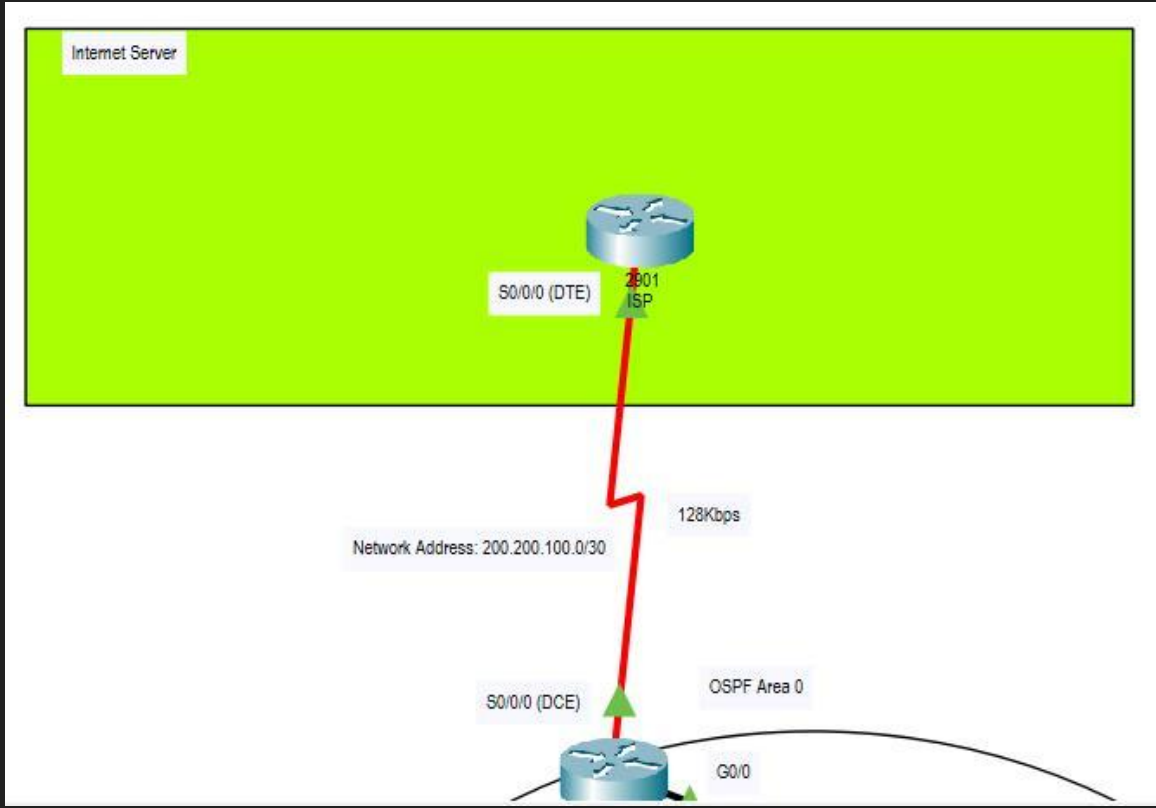
Company XYZ Logical Topology Diagram

- Internet Server
- Administration Building
- Remote Sales Office
- Main Building
  - Floor 1
  - Floor 2
  - Floor 3



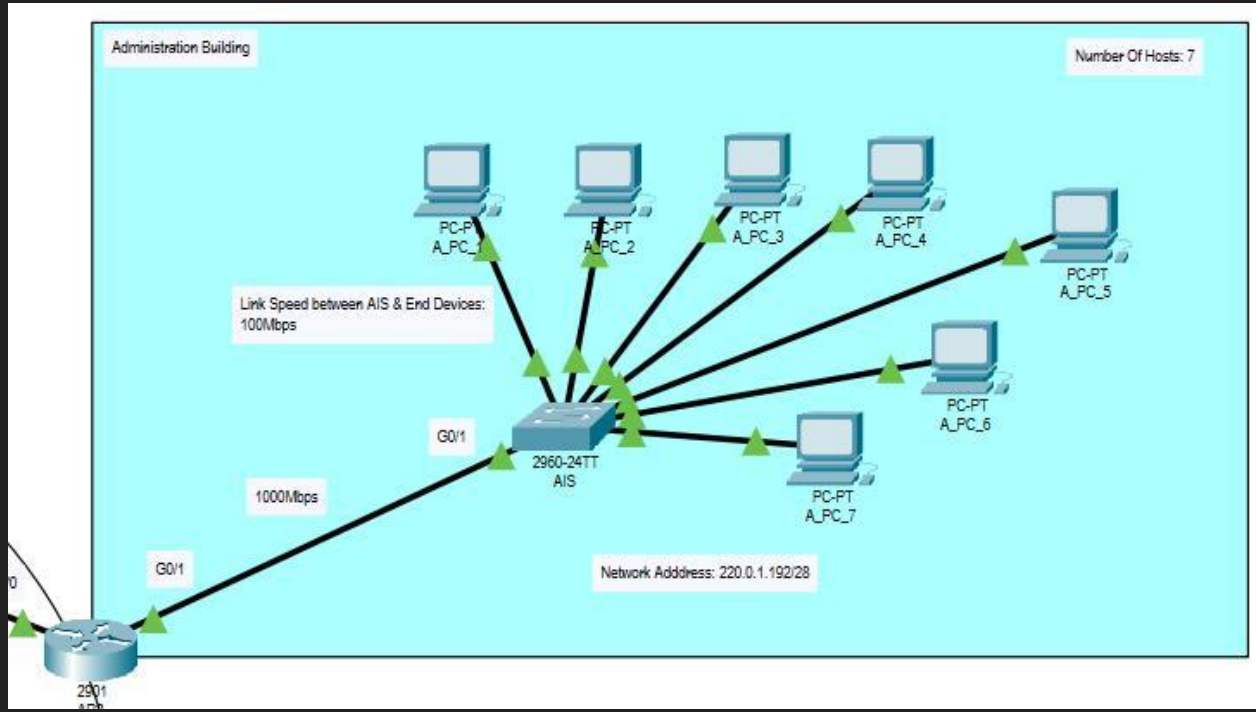
Internet Server

- Network Devices
  - ISP Router
  - AR1 (Edge Router)
- Hosts
  - Internet Server



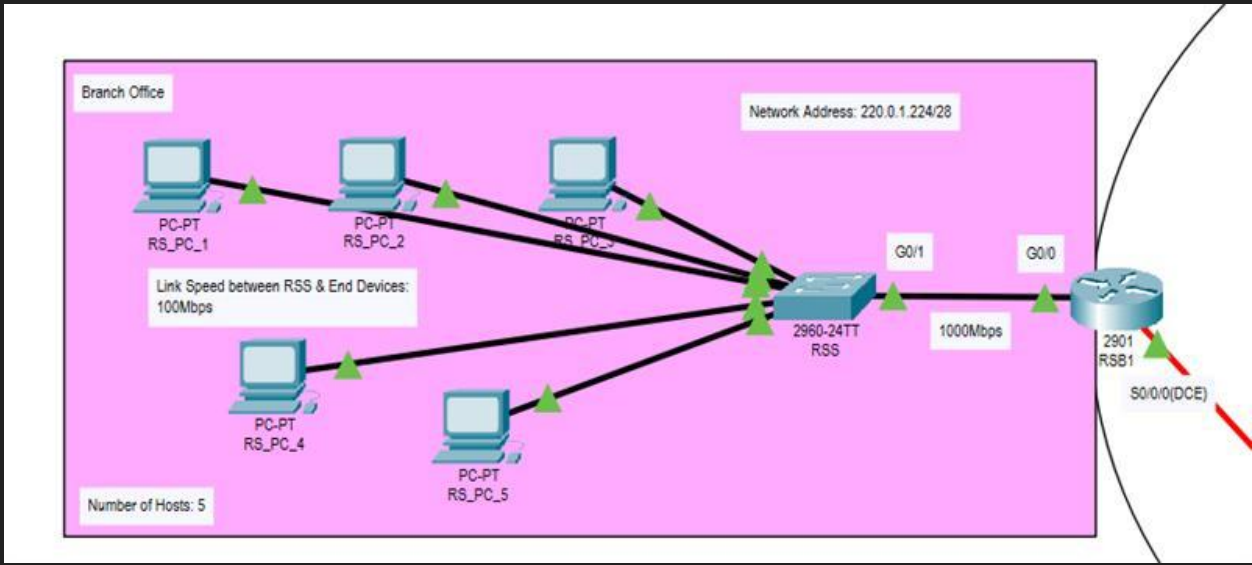
Administration Building

- Network Devices
  - AR2
  - AIS
- Hosts (7)
  - A_PC_X



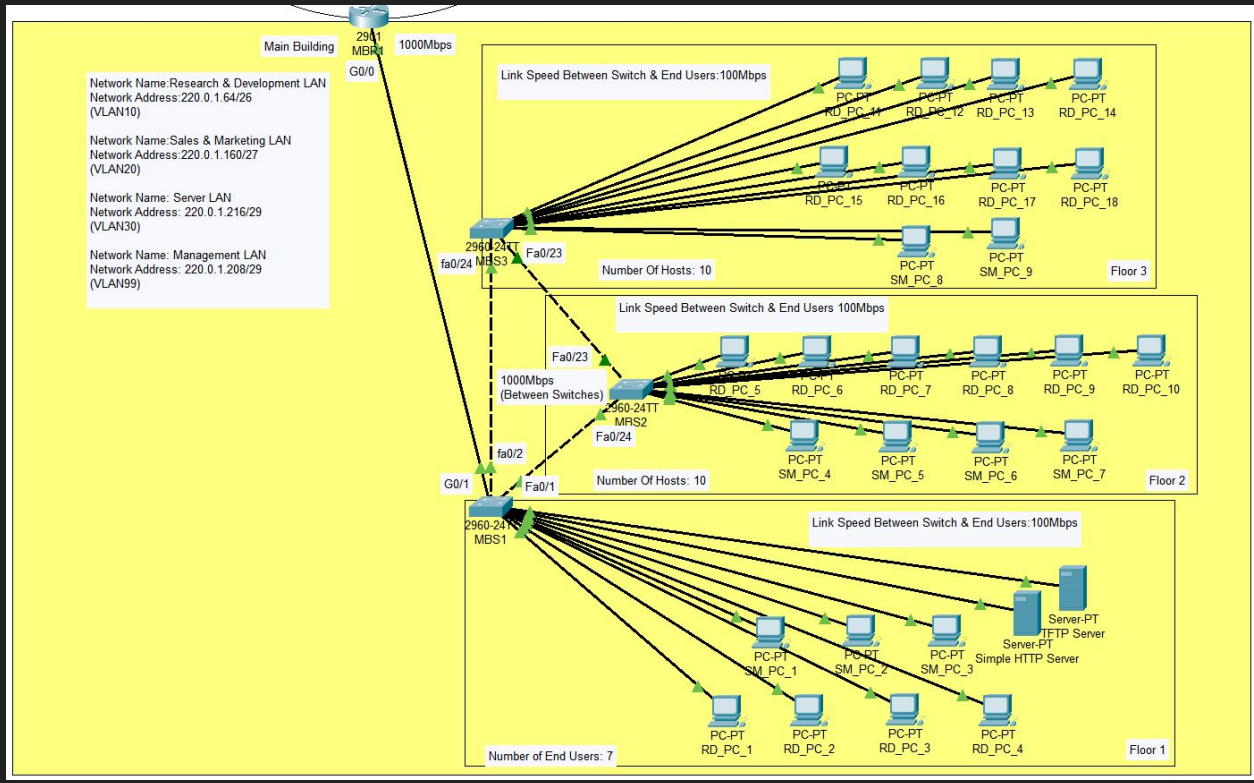
Branch Office

- Network Devices
  - RSB1
  - RSS
- Hosts (5)
  - RS_PC_X



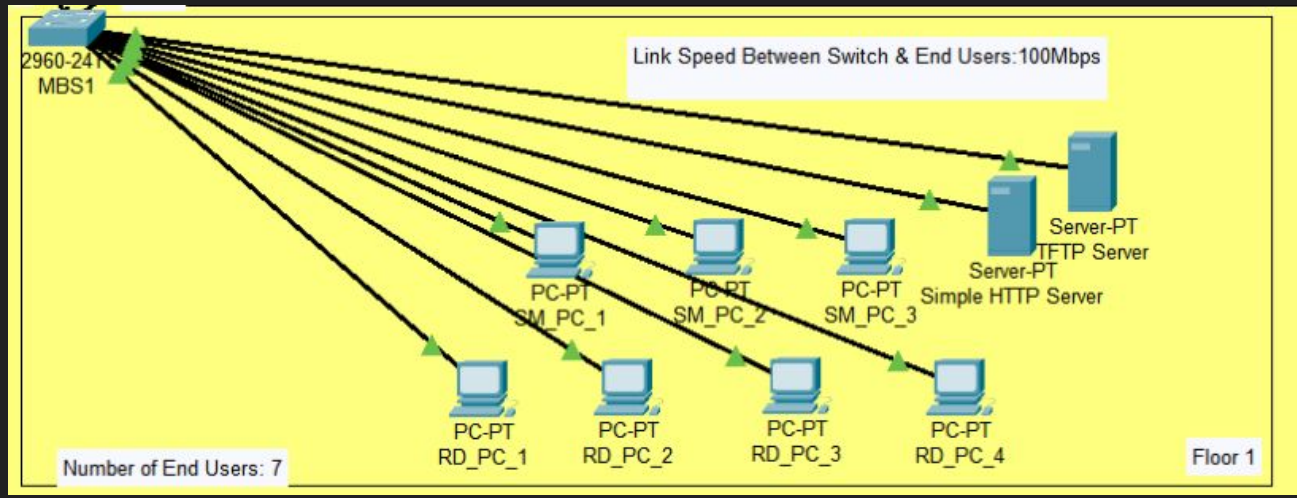
Main Building (Full View)

- Network Devices
  - MBR1
  - MBS1
  - MBS2
  - MBS3
- Hosts (27)
  - On Floors 1, 2, 3
- Servers (2)
  - HTTP
  - TFTP



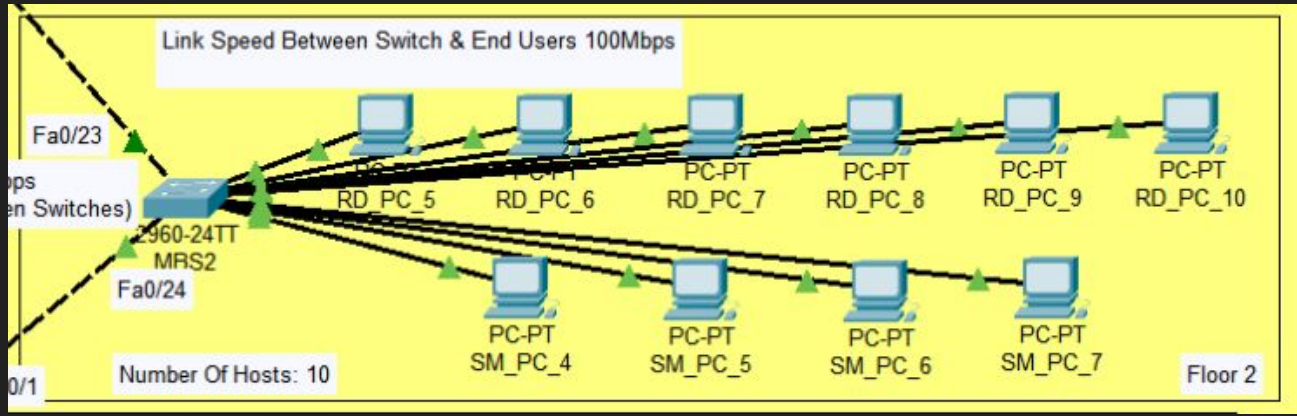
Main Building (Floor 1)

- Network Devices
  - MBS1
- Hosts (7)
  - RD_PC_X (4)
  - SM_PC_X (3)
- Servers (2)
  - HTTP
  - TFTP



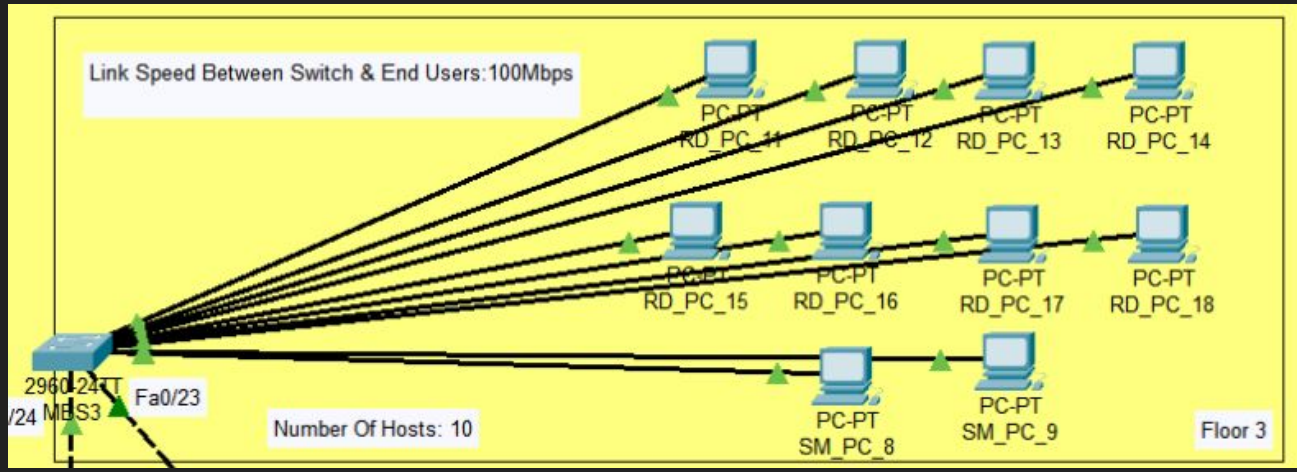
Main Building (Floor 2)

- Network Devices
  - MBS2
- Hosts (10)
  - RD_PC_X (6)
  - SM_PC_X (4)



Main Building (Floor 3)

- Network Devices
  - MBS3
- Hosts (10)
  - RD_PC_X (8)
  - SM_PC_X (2)



IP Addressing Scheme

IP Addressing Scheme

Company XYZ required the usage of VLSM in their IP addressing scheme.

- Class C 220.0.1.0/24 Network Block Requirements
  - Maximize available subnets (i.e. remaining subnets)
  - For PCs, Servers, Router Interface, Switch Virtual Interface, etc.
- 200.200.100.0/30 ISP Network
  - Used between edge router and ISP router
- Loopback Networks
  - 1.1.1.1/32, 2.2.2.2/32
    - Used in OSPF Routing Process

The 220.0.1.0/24 **block**

- 8 Subnets Required
  - LANs
  - P2P Links
  - VLANs
- 3 Available Subnets
  - Future use
  - Replanning

Number of host addresses required	Network Address (Host Range)	Subnet Mask	Max Number of Host Possible	In Use	Network Name
36	220.0.1.64 (.65 **to** .126)	/26	62	✓	Research and Development LAN
18	220.0.1.160 (.161 **to** .190)	/27	30	✓	Sales and Marketing LAN
14	220.0.1.192 (.193 **to** .206)	/28	14	✓	Admin LAN
10	220.0.1.224 (.225 **to** .238)	/28	14	✓	Remote Sales LAN
2	220.0.1.252 (.253 **to** .254)	/30	2	✓	Branch-Main WAN
2	220.0.1.248 (.249 **to** .250)	/30	2	✓	Admin-Main WAN
2	220.0.1.240 (.241 **to** .242)	/30	2	✓	Admin WAN
-	220.0.1.0 (.1 **to** .62)	/26	62	X	Unused
-	220.0.1.128 (.129 **to** .158)	/27	30	X	Unused
3	220.0.1.208 (.209 **to** .215)	/29	6	✓	Management LAN
5	220.0.1.216 (.217 **to** .223)	/29	6	✓	Server LAN
-	220.0.1.244 (.245 **to** .246)	/30	2	X	Unused

Other IP Addresses:

- 200.200.100.0/30
  - Edge to ISP
- Loopback Address
  - 1.1.1.1/32, etc.
  - OSPF RID

Other IP Address Ranges

Number of host addresses required	Network Address (Host Range)	Subnet Mask	Max Number of Host Possible	In Use	Network Name
2	200.200.100.0	/30	2	✓	Admin-ISP WAN
1	1.1.1.1	/32	1	✓	RSB1 OSPF RID
1	2.1.1.1	/32	1	✓	AR1 OSPF RID
1	2.2.2.2	/32	1	✓	AR2 OSPF RID
1	3.3.3.3	/32	1	✓	MBR1 OSPF RID

IP Addressing Scheme

Router Interfaces

- First available IP address in the subnet

Switch Virtual Interface

- Last available IP address in the subnet (AIS, RSS)
- Second available IP address in the subnet onwards (Main Building)

Host IP Address Scheme

- Second available IP address in the subnet onwards

Routing Protocols (L3)

Routing Protocols

Routing Protocol Table

Routing Protocol	Property 1 (Resource)	Property 2 (Convergence)	Property 3 (Scalability)	Property 4 (Use of VLSM)	Property 5 (Cisco Proprietary)	Total
RIPv1	Low resource usage (+)	Low convergence Speed (-)	Low (-) Maximum hop count of 15	Classful (-)	Vendor Neutral (+)	2
RIPv2	Low resource usage (+)	Low convergence Speed (-)	Low (-) Maximum hop count of 15	Classless (+)	Vendor Neutral (+)	3
OSPF	High Resource Usage (-)	High convergence Speed (+)	High Scalability with OSPF areas (+)	Classless (+)	Vendor Neutral (+)	4
EIGRP	High Resource Usage (-)	High convergence Speed with Feasible Successor (+)	High Scalability (+)	Classless (+)	Cisco Proprietary (-)	3
BGP	Low Resource Usage (+)	Low convergence Speed (-)	High Scalability (+)	Classless (+)	Cisco Proprietary (-)	3

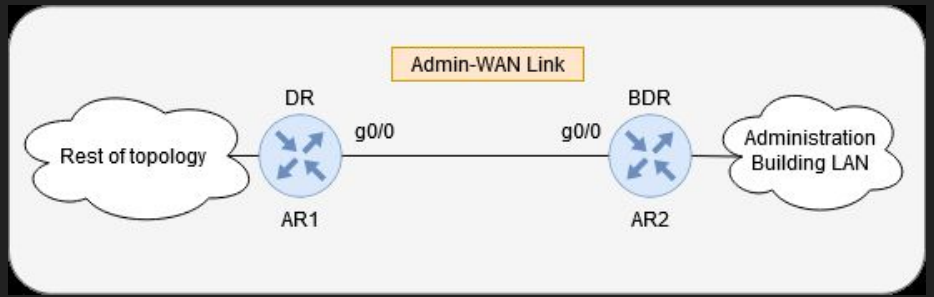
OSPF Requirements

- OSPF process id 50
- All routers in a single area, Area 0
- Loopback address used on each router
- Loopback address used as the OSPF router ID
- There is one multi-access network in the Administration building and a DR would need to be elected
- Determine whether to use Router ID or interface priority to determine the DR in the building

OSPF Considerations

Deciding between using Router ID (RID) or the highest loopback interface priority for DR Election:

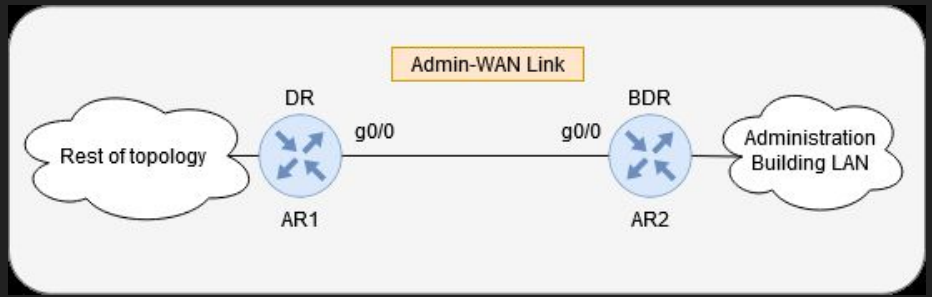
- Loopback addresses are used to determine RID
- DR Election considers highest loopback interface priority if implicitly defined
- To make sure the DR election stays consistent, we will use the highest loopback interface priority for DR election.



OSPF Considerations

Deciding which router will be the DR in the Main Building:

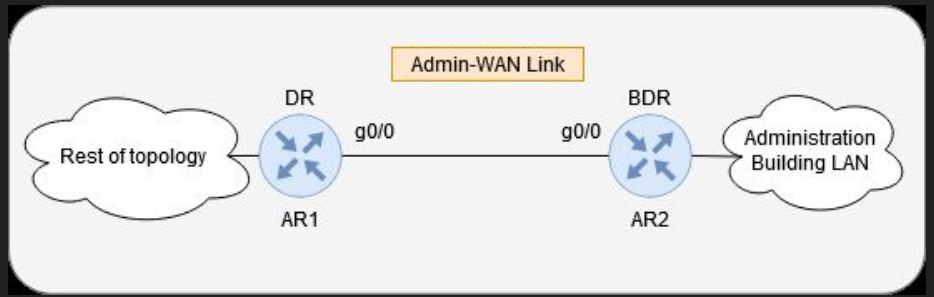
- Networks shared by AR1's OSPF process originates from rest of topology
- AR1 is elected as the DR as AR1 (intf. pri. 69)
- All OSPF Updates sent to DR before forwarding it to rest of multi-access network
- Saves bandwidth in the long run if more routers are added between AR1 & AR2



OSPF Considerations

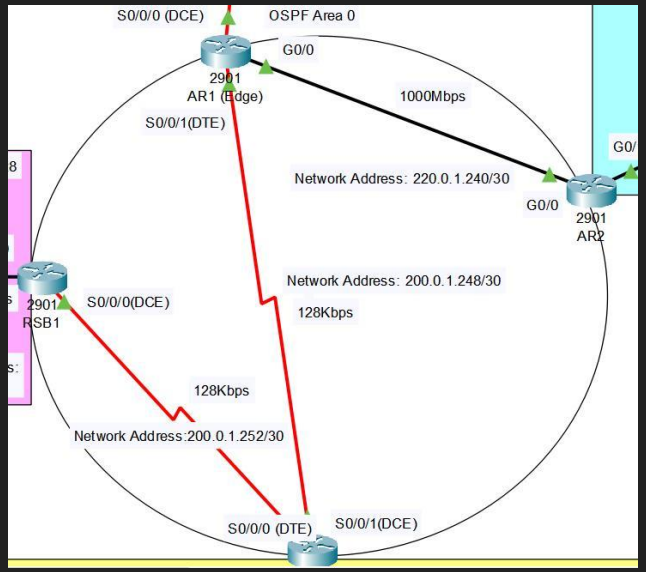
DR Election in XYZ's Network Topology:

- DR Election only occurs in multi-access network (i.e. ethernet links)
- Rest of the point-to-point connection between routers use serial link
  - Do not need to go through DR election
- Therefore, DR Election occurs only in XYZ's Admin-WAN Link.



OSPF Configurations

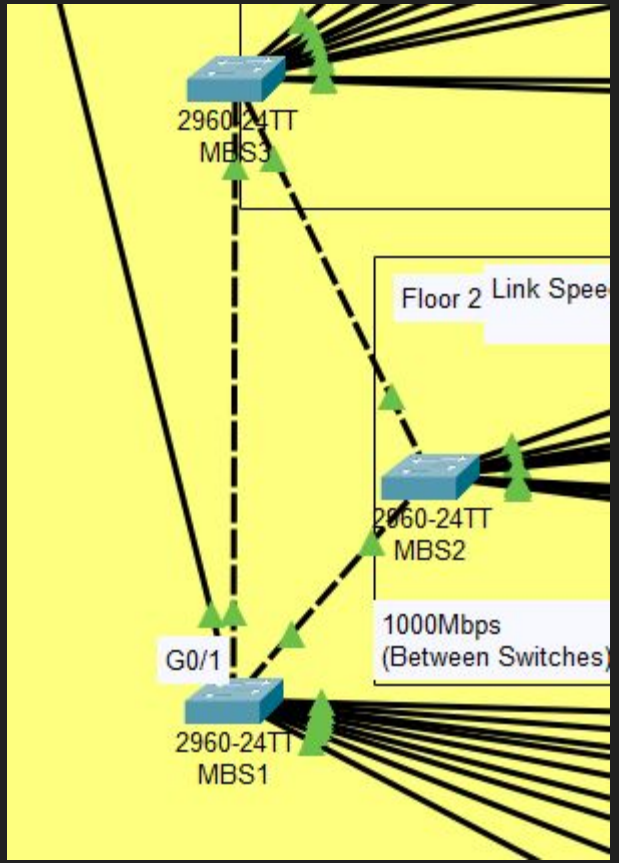
- Router ID Process of **50**
- Area 0
- Propagate all directly connected networks
- Passive-interface on interfaces facing the LAN
- Propagate default route on AR1 (Edge Router)



Switch Protocols (L2)

Switch Protocols Applied

- Virtual LANs (VLANs)
- Virtual Trunking Protocol
- Spanning Tree Protocol



Detailed Switch Table

Switch Name	Model	# of Ports	Location	IP Address	Gateway	Management VLAN	VTP Mode	VTP Domain	STP Root
MBS1	2960-24TT	24	Main Building Floor 1	220.0.1.210	220.0.1.209	99	Server	MBS1	VLAN 30
MBS2	2960-24TT	24	Main Building Floor 2	220.0.1.211	220.0.1.209	99	Client	MBS1	VLAN 20
MBS3	2960-24TT	24	Main Building Floor 3	220.0.1.212	220.0.1.209	99	Client	MBS1	VLAN 10
AIS	2960-24TT	24	Administration Building	220.0.1.206	220.0.1.193	1	-	-	-
RSS	2960-24TT	24	Branch Office	220.0.1.238	220.0.1.225	1	-	-	-

VLAN Requirements

- Two workgroups, dispersed randomly across three floors
- Server LAN on Floor 1
- Able to accomodate 100% growth
- Switches are connected in a loop; Network Redundancy is expected
- Spanning Tree Root to be placed in optimal position

VLAN

VLAN Table

VLAN Number	LAN Name
10	Research & Development LAN
20	Sales & Marketing LAN
30	Server LAN
60	Unused LAN (For unused ports)
99	Management LAN

VLAN 60 - Unused LAN

- Used for virtual storage of unused switch ports
- Ensures separation from production VLANs
- Switch ports will also be administratively down
- Considered to be a common security practice

VLAN 99 - Management LAN

- Specialised VLAN for Switch Virtual Interfaces (SVI)
- Allows for remote configuration of switches
- Ensures separation from production VLANs
- Considered to be a common security practice

Access & Trunking Ports

Detailed Switch Table B

Switch Name	# of Ports	Access Ports	Trunk Ports
MBS1	24	Fa0/3, Fa0/4, Fa0/5, Fa0/6, Fa0/7, Fa0/8, Fa0/9, Fa0/10, Fa0/11, Fa0/12, Fa0/13, Fa0/14, Fa0/15, Fa0/16, Fa0/17, Fa0/18, Fa0/19, Fa0/20, Fa0/21, Fa0/22, Fa0/23, Fa0/24, Gig0/2	Fa0/1, Fa0/2, Gig0/1
MBS2	24	Fa0/1, Fa0/2, Fa0/3, Fa0/4, Fa0/5, Fa0/6, Fa0/7, Fa0/8, Fa0/9, Fa0/10, Fa0/11, Fa0/12, Fa0/13, Fa0/14, Fa0/15, Fa0/16, Fa0/17, Fa0/18, Fa0/19, Fa0/20, Fa0/21, Fa0/22	Fa0/23, Fa0/24, Gig0/1, Gig0/2
MBS3	24	Fa0/1, Fa0/2, Fa0/3, Fa0/4, Fa0/5, Fa0/6, Fa0/7, Fa0/8, Fa0/9, Fa0/10, Fa0/11, Fa0/12, Fa0/13, Fa0/14, Fa0/15, Fa0/16, Fa0/17, Fa0/18, Fa0/19, Fa0/20, Fa0/21, Fa0/22	Fa0/23, Fa0/24, Gig0/1, Gig0/2

VLAN Trunking Protocol (VTP)

- Cisco Proprietary Protocol
- Propagates VLANs across the defined domain
- Advertises over 802.1q trunks (and ISL trunks)
- Comes in three modes: Server, Client, and Transparent
- Allows for ease of configuration and consistency

VTP Configuration

- VTP Version 2
- VTP Operating Mode: MBS1 (Server), MBS2 (Client), MBS3 (Client)
- VTP Domain Name: MainBuilding
- VTP Pruning Mode: Disabled
- VLANs Propagated: R&D, S&M, Server, Unused, Management

Spanning Tree Protocol (STP)

- Used for building loop-free logical topologies
- Prevents bridge loops and broadcast storms
- Operates by designating a root bridge
- Monitors the network topology constantly (using BPDUs)
- Catalogues every link between switches; especially redundant links
- All switches selects their best connection to designated root bridge
- All other redundant links are blocked

STP Configuration

- Rapid PVST+ configured as STP implementation
- Implemented on a Per-VLAN basis
- Bridge Priority manually configured for all VLANs

Spanning-tree bridge table summary

Spanning-tree Process	Primary Root Bridge	Secondary Root Bridge
VLAN 10	MBS3	MBS2
VLAN 20	MBS2	MBS1
VLAN 30	MBS1	MBS3

Root Bridge Placement

- Determined based on number of hosts connected to each VLAN on the switches
- Example: 8 R&D staffers connected to MBS3, 4 R&D staffers connected to MBS1. MBS3 will be the root bridge for the R&D VLAN.
- This criteria also ensures each switch is a root bridge for a different VLAN
- Provides load-balancing and better bandwidth efficiency

Additional STP Features

- PortFast
  - Usually configured on edge ports (for end devices)
  - Immediately transitions the port to forwarding state (skips listening & learning states)
  - Enables end devices to start forwarding traffic as soon as possible
- BPDUGuard
  - Used to protect against BPDU-related attacks
  - Should enabled on edge ports (ports which should not receive BPDUs)
  - Commonly-used together with PortFast
  - Disables the port (Error-Disabled state) upon receiving a BPDU

Physical Layer (L1)

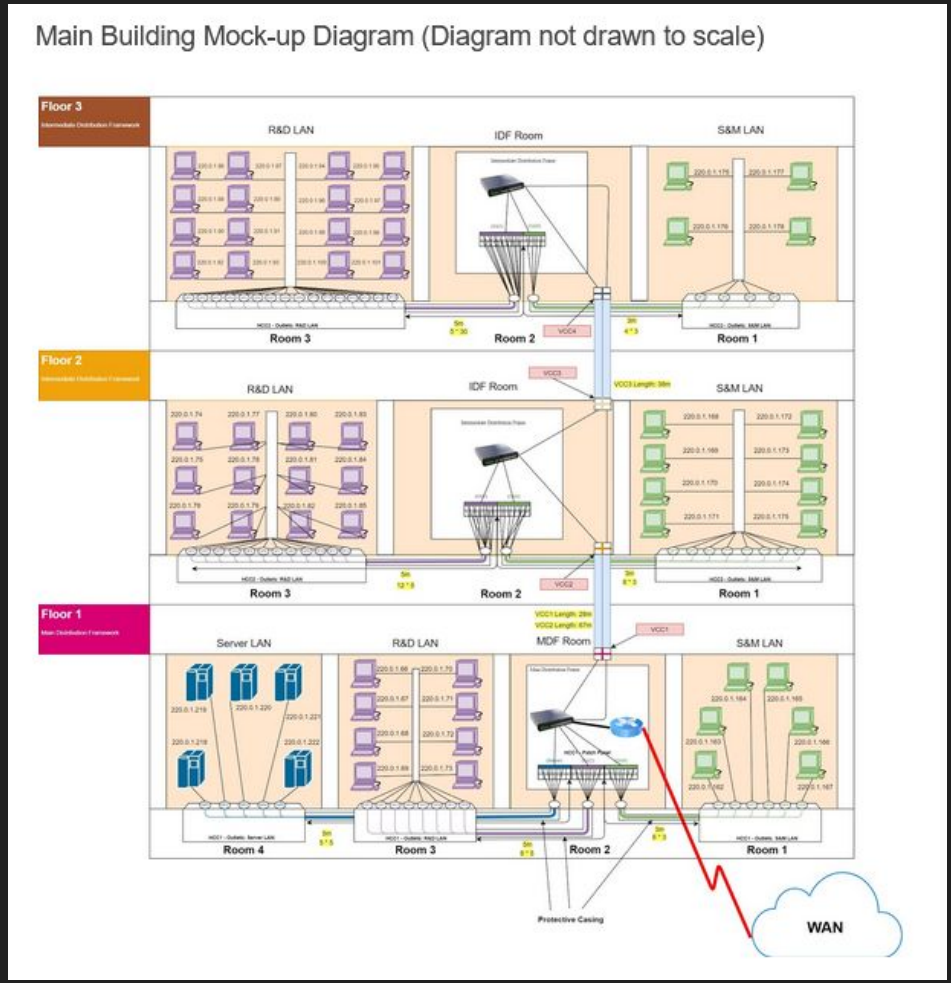
Physical Layer

Based on the company’s requirements, we have come up with a physical layer design for the Main Building. This consists of:

- Physical Layer diagrams
- Labeling convention
- Equipment needed
- Total cost of the equipment

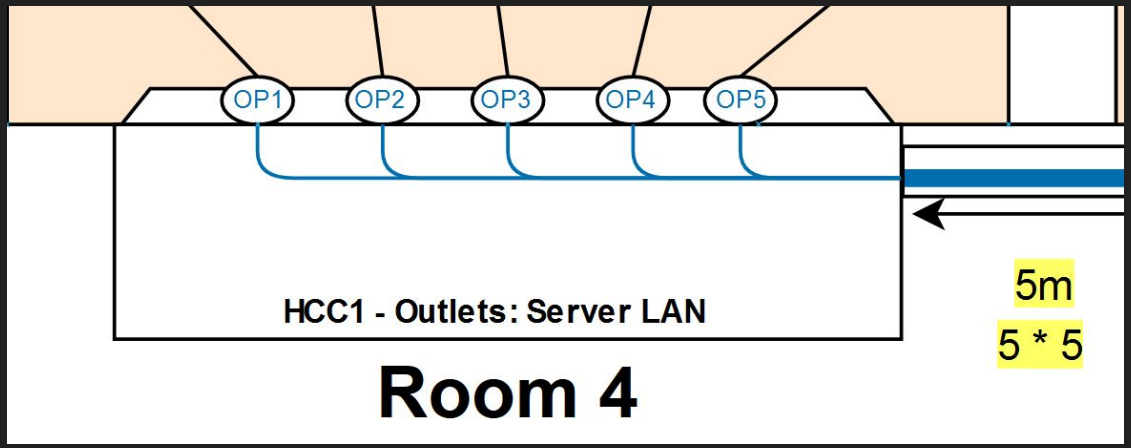
Main Building Mock-up Diagram

- X-rayed Front-View
- Three Floors
- Rooms for each workgroup
- 100% growth of hosts factored in
- Backbone cabling featured



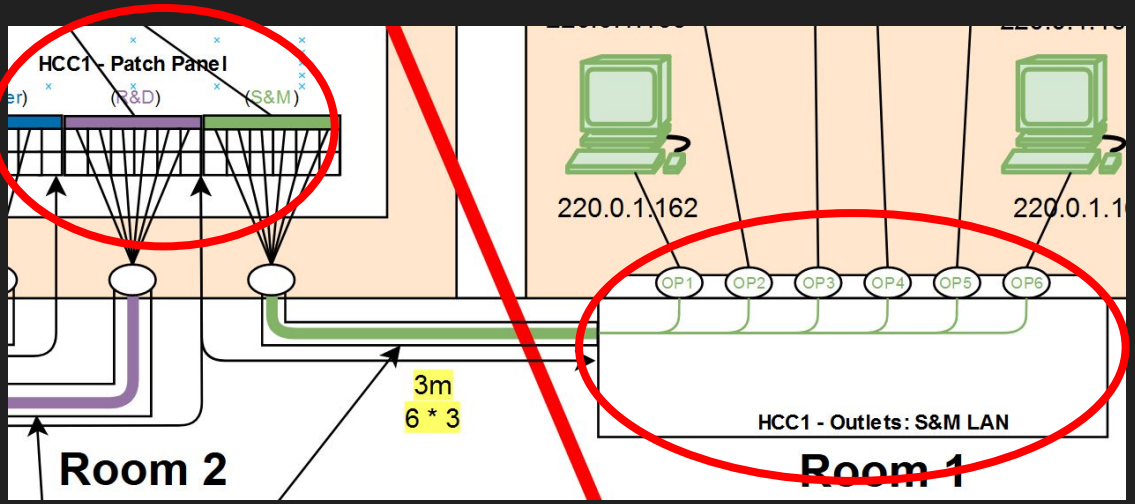
Sample Horizontal Cross-Connect (Outlet)

- All ports labeled on the diagram
- Horizontal Cross-Connect labeled accordingly (elaborated further)
- Room Number
- Cable lengths
- No. of Cable runs



Sample Horizontal Cross-Cabling

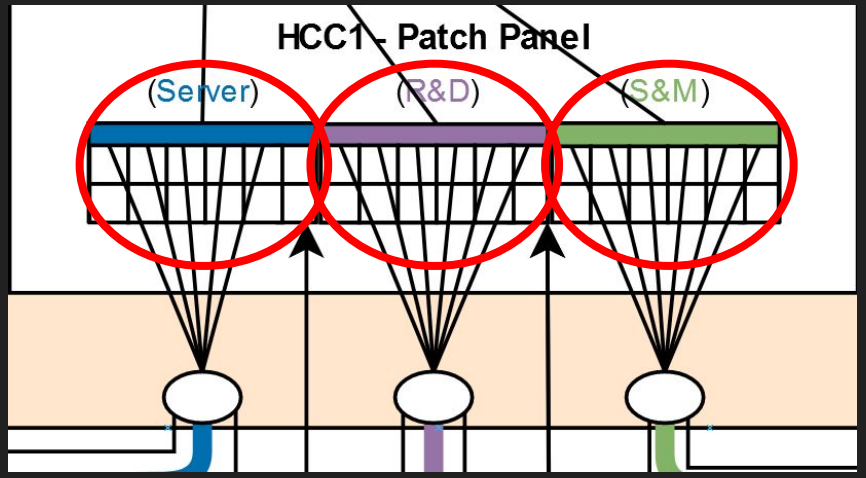
- HCC - Outlets connect to HCC - Patch Panel
- Cabling runs under raised flooring, protected in PVC pipes.
- Number of cables & cable length listed



Sample Horizontal Cross-Connect (Patch Panel)

- Patch Panel segmentation is labeled
- Connections are color-coded
- Number of Cables match exactly

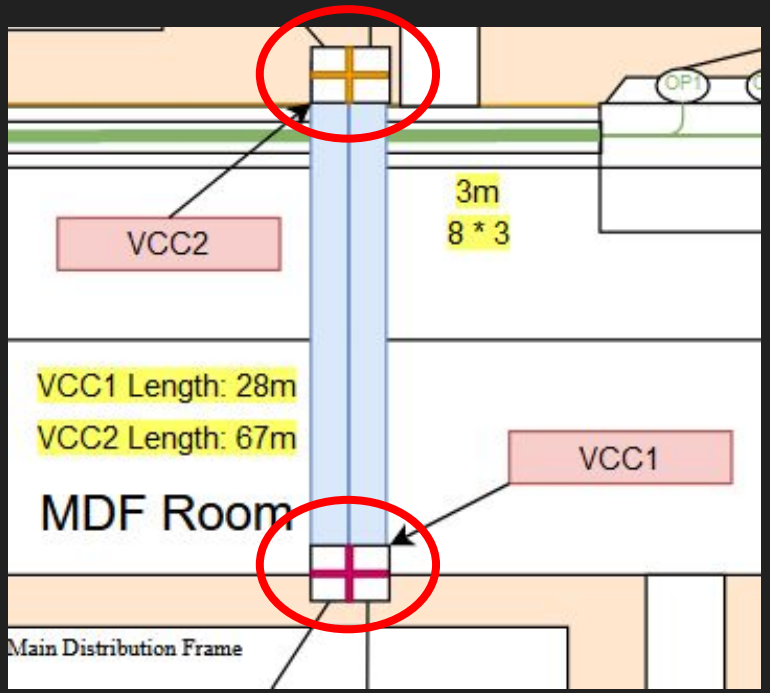
Note: Patch Panel Labeling is on a separate diagram



Sample Vertical Cross-Connect (Patch Panel)

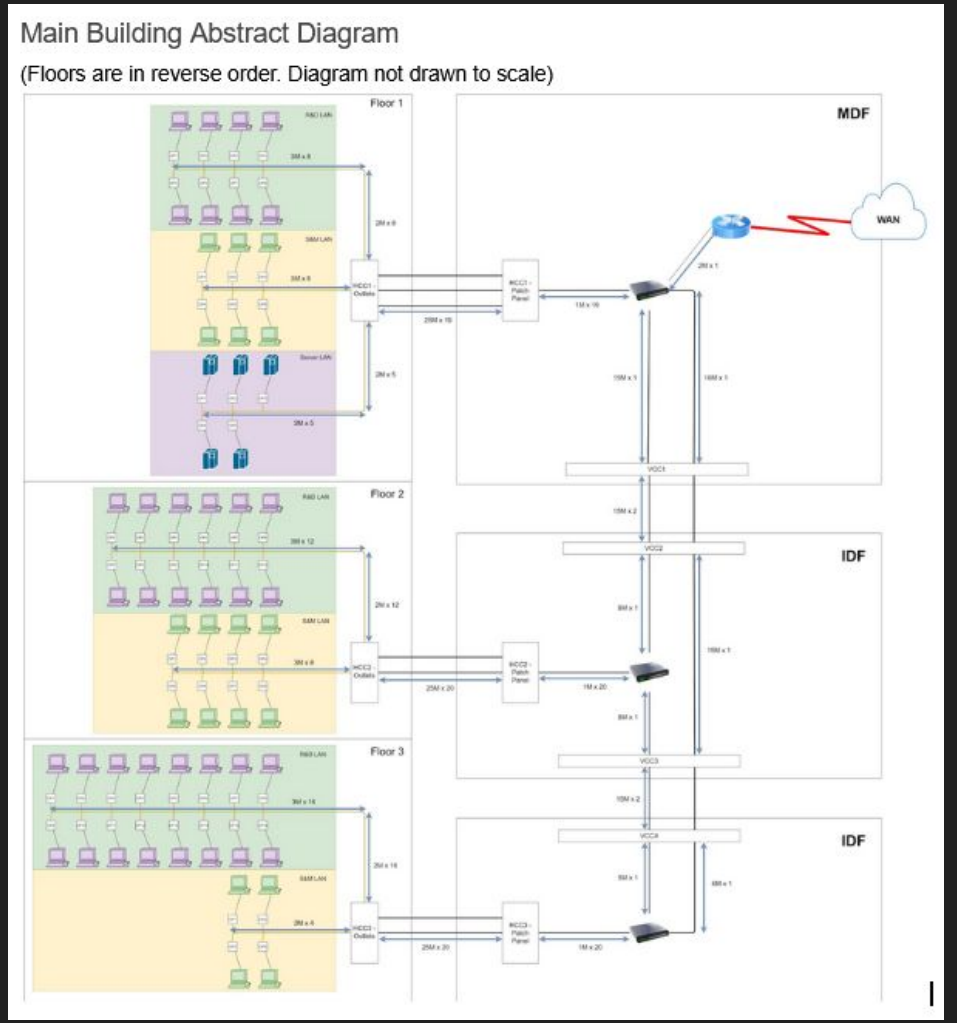
- Patch Panels are color-coded
- Connections are protected by PVC pipes
- Number of cables & cable length listed

Note: Patch Panel Labeling is on a separate diagram



Main Building Abstract Diagram

- Logical View
- Mirrors data from Mock-Up Diagram
- Added focus on connections
- Clearer labeling



HCC - Patch Panel Labeling

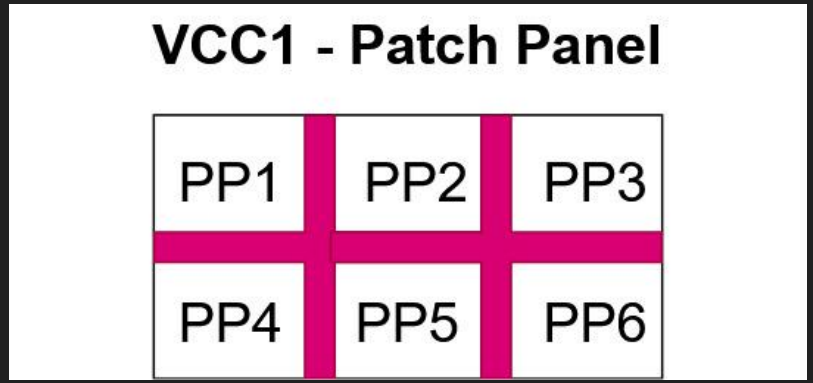
- Labeling Scheme Sample: ‘A’ - PP’B’
- ‘A’ will be short-form of the LAN segment (S, RD, or SM)
- ’B’ will be number of the port (number starts from 1 for each segment)

HCC1 - Patch Panel

(Server)						(R&D)						(S&M)					
S-PP1	S-PP2	S-PP3	S-PP4	S-PP5	S-PP6	RD-PP1	RD-PP2	RD-PP3	RD-PP4	RD-PP5	RD-PP6	SM-PP1	SM-PP2	SM-PP3	SM-PP4	SM-PP5	SM-PP6
S-PP7	S-PP8	S-PP9	S-PP10	S-PP11	S-PP12	RD-PP7	RD-PP8	RD-PP9	RD-PP10	RD-PP11	RD-PP12	SM-PP7	SM-PP8	SM-PP9	SM-PP10	SM-PP11	SM-PP12

VCC - Patch Panel Labeling

- Labeling Scheme Sample: Floor’A’xFloor’B’-PP’C’
- ‘A’ will be the floor which the Patch Panel is on
- ’B’ will be the floor with the connected Patch Panel
- ’C’ will be the number of the port



Main Building Cabling Details

- Ethernet Category used: Category 8.2 (throughout the building)
- Pinout Type: Straight-Through (for all connections)
- Total Length of Cabling: 1928 meters
- Cabling Types used: Solid Core and Stranded (Ethernet)

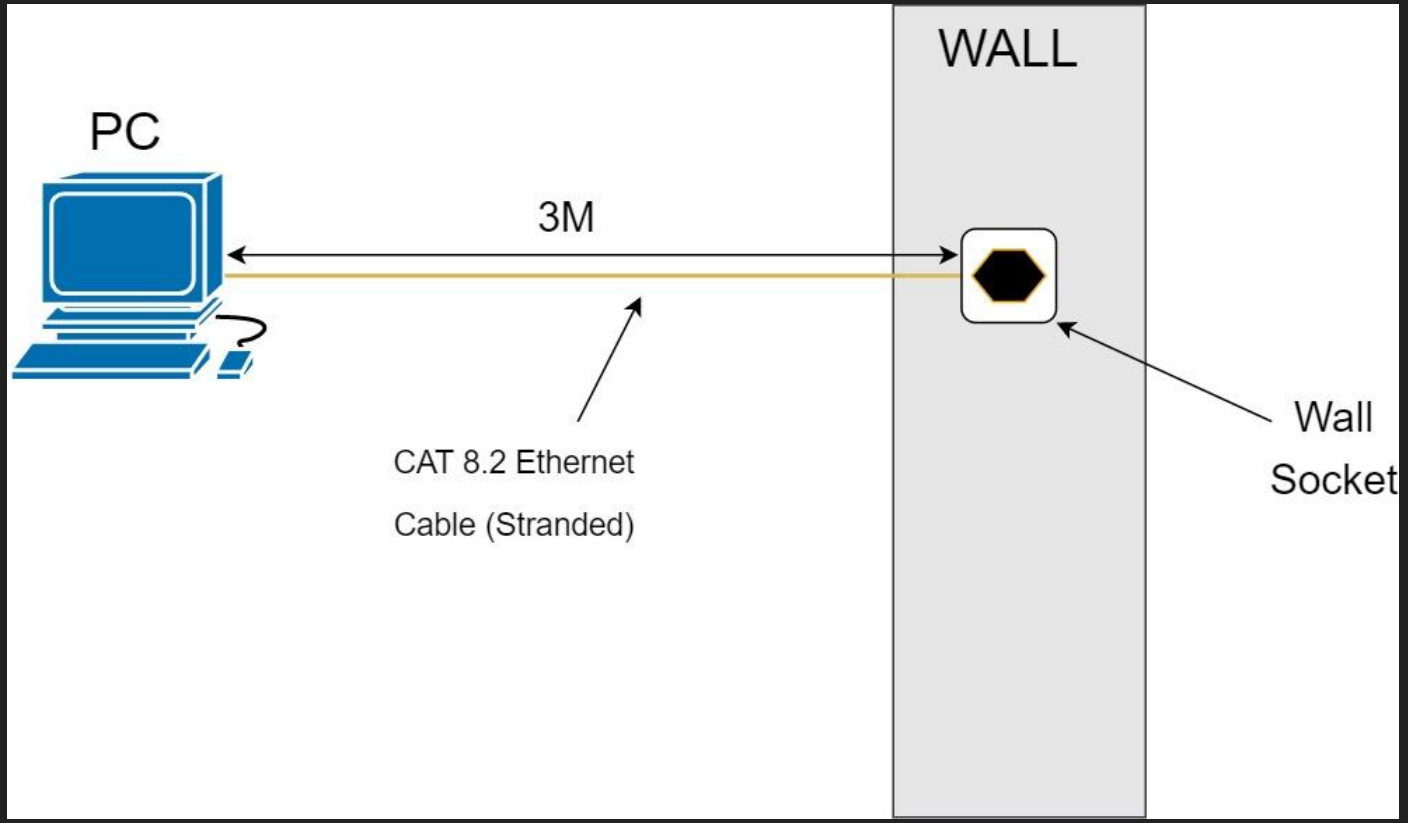
Solid Core Ethernet Cabling

- Used for backbone and horizontal cable runs
- Should not be overly bent, flexed or twisted
- Lower attenuation than stranded cabling
- Optimal choice for long-term installations

Stranded Ethernet Cabling

- Used for shorter connections (between end devices and outlets, etc)
- Much more flexible than solid core cabling
- Higher attenuation than solid core cabling
- Optimally kept to minimum length

Sample PC - Outlet Diagram



Physical Layer Equipment

- Represent the network
- Sourced for the various equipment necessary for establishing it
- Source the pricing of the equipment from various online retailers
- Took the average price calculated from the various sources

Main Building Cable Parts

Equipment Type	Model No.	Qty	Cost (SGD)
Ethernet Cable (Solid core)	CAT8.2	10 Spool (1 spool = 150m, 10 spool needed - 1475m)	$9,500
Ethernet Cable (Stranded)	CAT8	2 Spool (1 spool = 305m, 2 spool needed - 393m)	$1,787
RJ45 Jacket	CAT8.2	124 (1 box = 100pcs, 2 boxes needed)	$47.60
RJ45 Wall Socket	CAT8.2	59	$2,082.70
PVC Pipe	Georg Fischer	800 pipes, 2m long x 60mm OD, 4.5mm Wall Thickness	$36,410
Cable Tie	Heavy Duty 26 Inch	91 (1 packet = 50pcs, 2 packets needed)	$44 Total: $49,871.30

Main Building Cable Parts

- All cables to be created manually
- Total cable equipment to be purchased based on calculations
- Prices were sourced from various online retailers
- Items selected based on a balance of:
  - Highest Quality
  - Cheapest Price
  - Minimal wastage

Main Building Cable Parts

- Solid Core cable spool for creation of backbone cables
- Stranded cable spool for creation of flexible cables
- RJ45 jackets for establishing cable connectors
- RJ45 wall sockets for cable termination (connect to backbone cabling)
- PVC pipes to serve as protection for backbone cabling
- Cable ties to ensure proper cable management

Main Building Floor 1, Room 2, MDF

Equipment Type	Model No.	Qty	No. /Type Ports	Description/ Function	Cost (SGD)
Router	2901	1	2 integrated 10/100/1000 Ethernet ports (Gig)	Forwards data packets between computer networks.	$1,900
Serial Network Module	HWIC-2T	1	2 serial interfaces	Provide serial interfaces to the router	$200
Switch	2960-24TT	1	24 integrated 10/100 Ethernet ports (Fa), 2 integrated 100/1000 Ethernet ports (Gig)	Connects to other network devices and end devices.	$1,400
24 ports Patch Panel	WC351088	1	24 Ports, RJ45 and rear 110-type terminations.	Connects routing circuits for monitoring, interconnecting, and testing circuits in a convenient, flexible manner	$475
6 ports Patch Panel	Yankok [6 Port Patch Panel]	1	6 ports, RJ45 and rear 110-type terminations.	Connects routing circuits for monitoring, interconnecting, and testing circuits in a convenient, flexible manner	$15
Rack	QFR1924-45-BK	1	-	Rack to store/hold the network equipment and patch panel	$445 Total: $4,435

Main Building Floor 1, Room 2, MDF

Equipment Purchased:

- Rack
- Router, including serial interface
- Switches
- Patch Panels
- Prices above were all cross-referenced for the average price

Sample Main Building, IDF

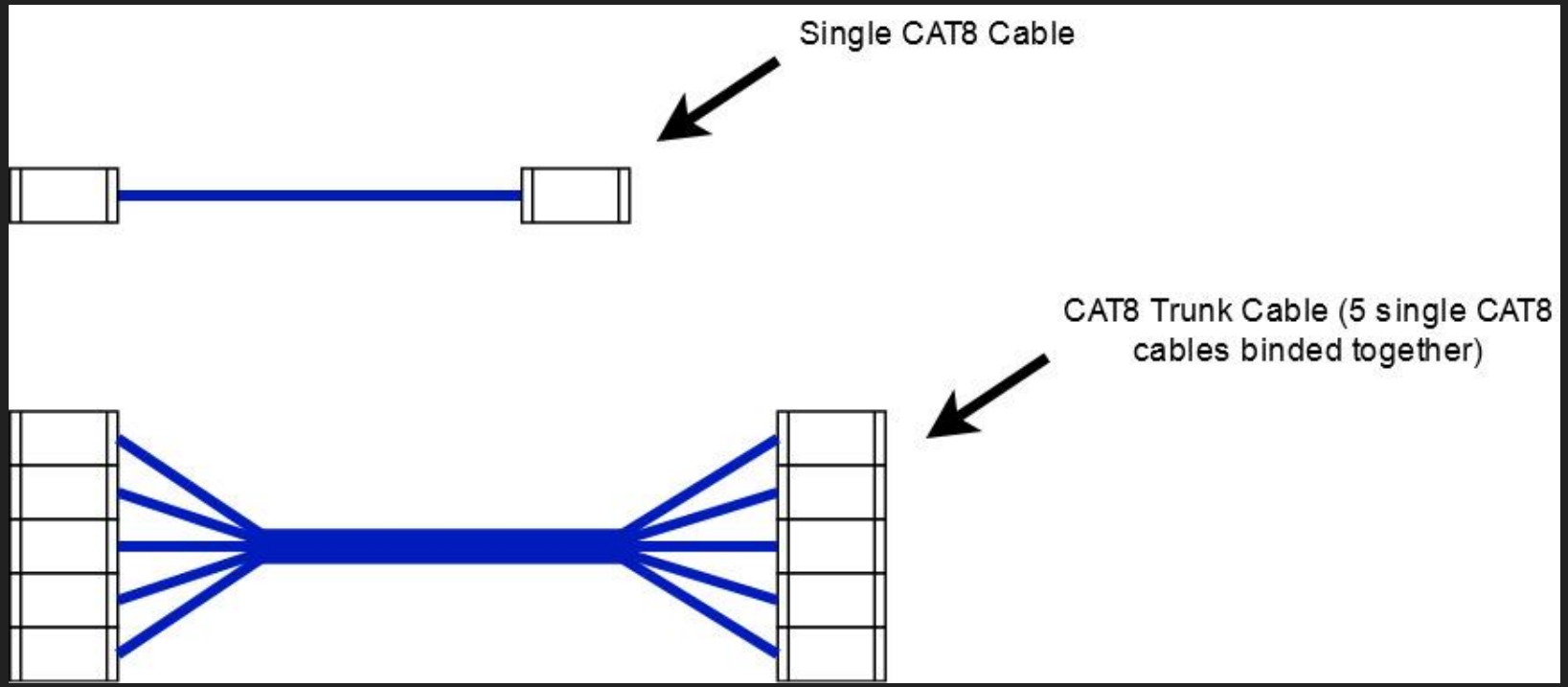
Equipment Type	Model No.	Qty	No. /Type Ports	Description/ Function	Cost
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6 ports Patch Panel	Yankok [6 Port Patch Panel]	1	6 ports, RJ45 and rear 110-type terminations.	Connects routing circuits for monitoring, interconnecting, and testing circuits in a convenient, flexible manner	$15
Rack	QFR1924-45-BK	1	-	Rack to store/hold the network equipment and patch panel	$445
					Total: $2,335

Sample Main Building IDF

Equipment Purchased:

- Rack
- Switches
- Patch Panels
- Prices above were all cross-referenced for the average price

Sample Trunk Cable Diagram



Sample Trunk Cable Diagram

- Diagram illustrates how a trunk cable is created for use
- No Cat 8 trunk cables readily available on market
- Must be custom-ordered from factories (exorbitant prices)
- Opted to create the trunk cables
  - Calculated and created a large number of Cat 8 cables
  - Binded multiple Cat 8 cables with heavy-duty cable ties
  - Each Cat 8 cable still labeled individually (after trunk cable label)

Testing & Troubleshooting

Testing & Troubleshooting

- Routing between Main Building VLANS
  - Ping Test
- All hosts connecting each other and HTTP Service
  - Ping Test/Traceroute to identify problems
- Single Trunk Link happens to fail
  - Which route will the packet travel next
- Trivial File Transfer Protocol (TFTP)
  - cisco packet tracer limitation

Main Building VLANS

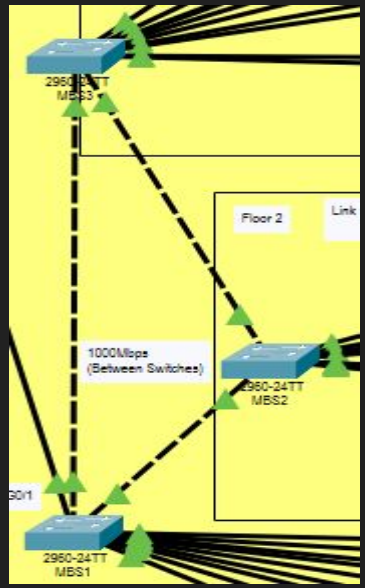
- Between Hosts in VLAN 10 and VLAN 20
  - Research Development LAN & Sales Marketing LAN
- Web access to web server VLAN 30 and VLAN 10
  - HTTP Server & Research Development LAN
- Web access to web server VLAN 30 and VLAN 20
  - HTTP Server & Sales Marketing LAN

Connectivity between all hosts and HTTP Server

- RD_PC_1 (220.0.1.66)
  - SM_PC_1 (220.0.1.162)
  - RS_PC_1 (220.0.1.226)
  - HTTP Server (220.0.1.194)
  - A_PC_1 (220.0.1.194)
- All devices Ping 5 different networks successfully

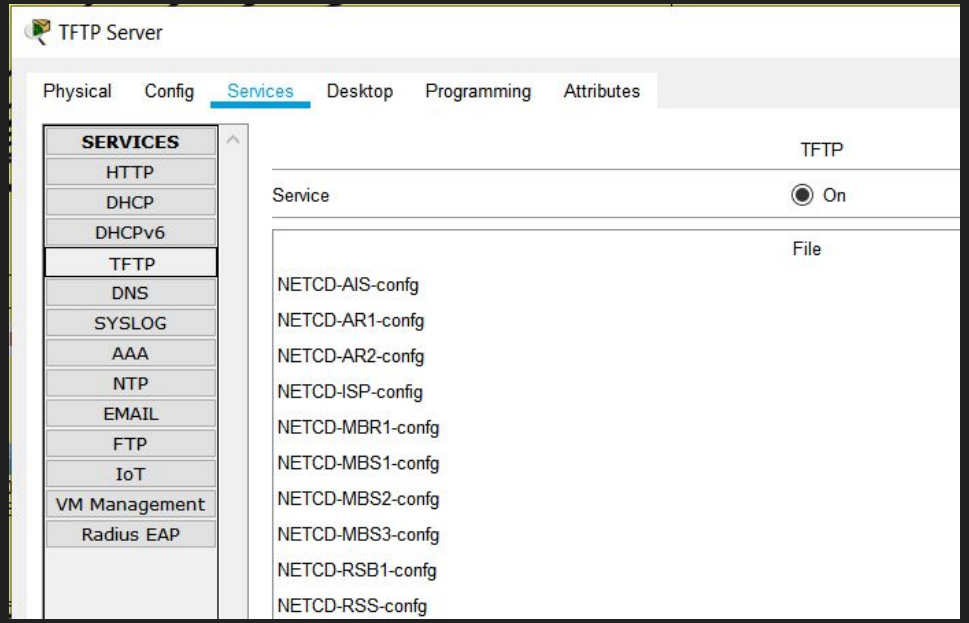
Single Trunk routes

- RD_PC_11 (VLAN 10, MBS3) to RD_PC_1 (VLAN 10, MBS1)
  - Trunk is UP
    - Route: RD_PC_11 → MBS3 → MBS1 → RD_PC1
  - Trunk is DOWN
    - Route: RD_PC_11 → MBS3 → MBS2 → MBS1 → RD_PC1
- Spanning Tree Process functions well in the event of a trunk link failure
  - Ensuring reliability and redundancy



TFTP

- Main Building Server LAN
- “Copy Running-Config” of all devices to tftp server
- Limitation of copying the ping/router table output to the tftp server

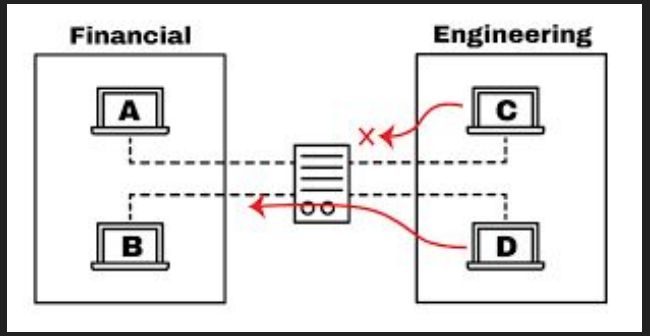


Network Recommendations

Network Recommendations

1. Access Control Lists
2. Etherchannel
3. Multilink
4. Hot Standby Routing Protocol
5. Implement Cloud Computing
6. IDS & IPS

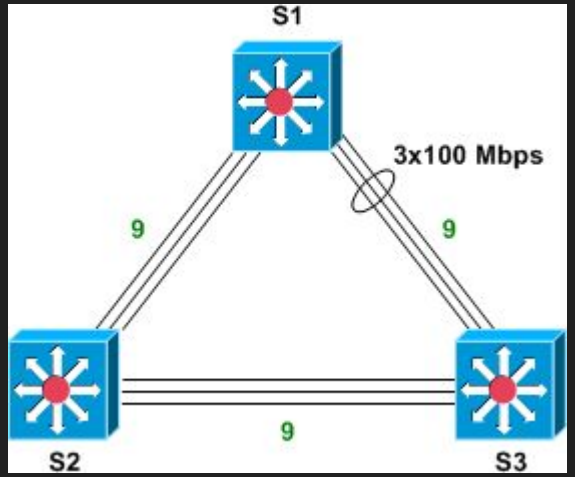
Access Control Lists

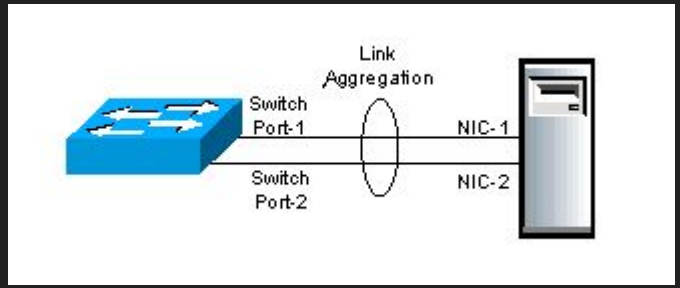


- Can be implemented on both router & switches
  - Interfaces
  - VTY Lines
- Filters network traffic based on predefined rules
- Can be used for implicit forwarding or denying of packets
- Able to filter based on upper-layer protocols (router only)
  - Extended ACLs

Etherchannel

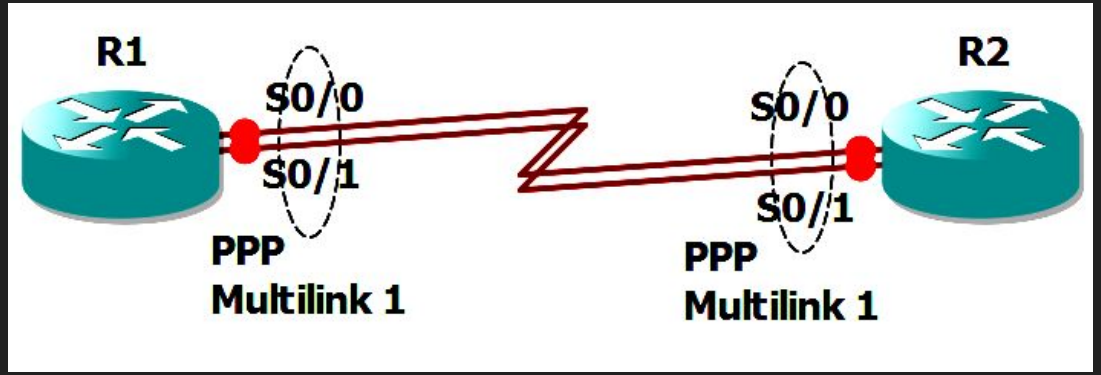
- Cisco Proprietary port link aggregation technology
- Allows for grouping of physical Ethernet links
- Forms a single logical Ethernet link
- Provides fault-tolerance
- Increased bandwidth
- Redundancy for HTTP Servers in Main Building
  - Ensures availability in case of a link failure





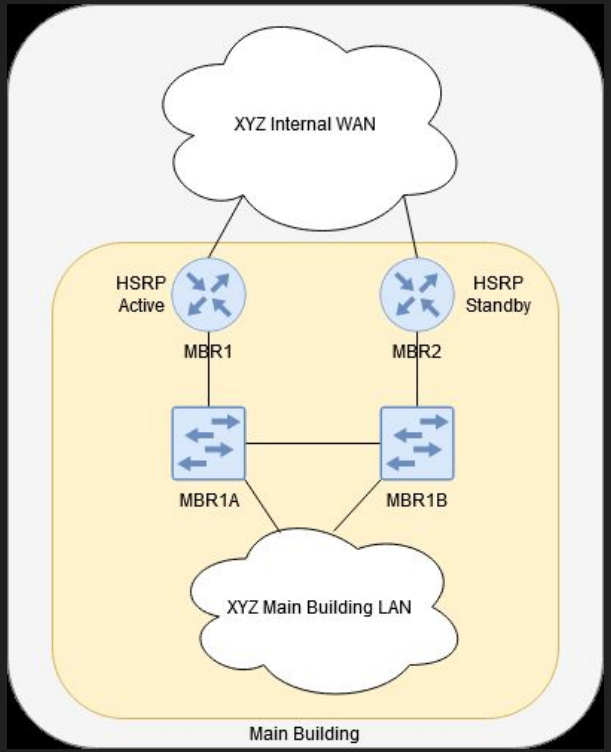
Multilink

- Similar to Etherchannel, but applied for serial connections
- Allows for grouping of physical serial connections
- Forms a single logical serial link
- Provides fault-tolerance
- Increased bandwidth



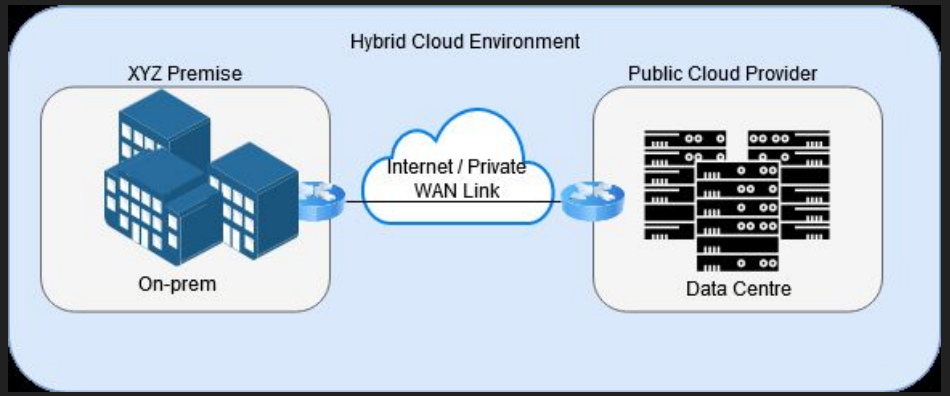
Hot Standby Routing Protocol

- Cisco Proprietary redundancy protocol
- Allows for grouping of multiple routers
  - One active router responsible for forwarding traffic
  - Other routers to serve as backups
  - When active router fails,
    - next designated standby router will take over
  - Per-VLAN load balancing
- Forms a single virtual router
- Establishes a fault-tolerant default gateway



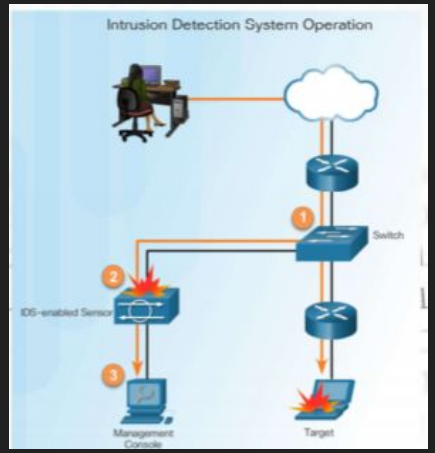
Implement Cloud Computing

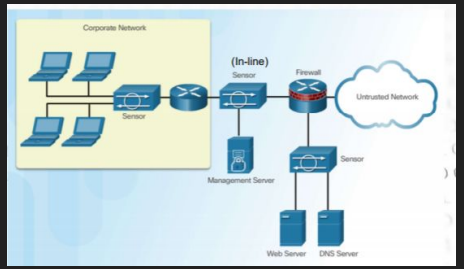
- Hybrid Cloud Environment
- Allows for flexibility & scalability if not enough servers
  - Variable Expense (Pay as you go)
  - Scales computing resources up & down depending on workload
- Provides redundancy if on-prem server fails / high server load
  - Ensures HTTP Server Availability
- AWS / Azure



Intrusion Detection System

- Intrusion Detection System (IDS)
  - Detects and logs suspicious traffic
  - Parallel
- Intrusion Prevention System (IPS)
  - Detects, log, and **filter** suspicious traffic
  - Inline





IDS & IPS can be placed on all of XYZ sites for additional **network security** measures.

Final Demonstration

Conclusion

- Successful completion of XYZ network design
- Application of network knowledge learnt from school
- Learnt how to design physical layer diagrams
  - Gained in-depth knowledge of various physical equipment (Patch Panels, Cabling type, etc)
- Improved research skills
  - Retrieve relevant information more effectively
  - Cross-referencing against multiple sources

End

References & Links

Web Images:

- https://packetlife.net/media/blog/attachments/282/layer2.png
- https://upload.wikimedia.org/wikipedia/commons/5/5b/Link_Aggregation1.JPG
- https://www.google.com/url?sa=i&url=https%3A%2F%2Fciscoarticleandcertification.blogspot.com%2F2017%2F02%2Fppp-multilink.html&psig=AOvVaw3PjHj7S92yVIa8gtNwF6nf&ust=1605245230085000&source=images&cd=vfe&ved=0CAIQjRxqFwoTCNiiy9Si_OwCFQAAAAAdAAAAABAD

Generated Physical Diagrams:

- https://drive.google.com/file/d/1_6wt29F2kM0BfR5liemQv5cjpk2_BPv9/view?usp=sharing
    - This is a google drive link to the draw.io file containing our physical layer diagrams. This is only accessible by email accounts within the “Ngee Ann Polytechnic” Group